Smart Irrigation Decision Support System Using Simulated Soil Moisture and Weather Data

Java-Based Rule-Driven Irrigation Simulation Project

CSA 0904 – Java Programming

Presented By
O. Bhanu prakash Reddy 192372075
M.Mahi reddy 192324082
K.Haneesh 192324084
Sri Hari 192372222
Antony Ajay 192524414

Guided By

Dr. Thalapathy Rajasekaran

Introduction

- Efficient irrigation is essential for sustainable agriculture, yet many decisions are still made manually without real-time environmental data.
- **Smart Irrigation Decision Support System** is a Java-based application that simulates soil moisture and weather conditions for demonstration purposes.
- The system analyzes simulated environmental data using predefined decision rules to recommend irrigation actions.
- No real IoT sensors are used; simulated data is generated to demonstrate the complete decision-making workflow.
- The system provides recommendations, alerts, and historical reports to support informed irrigation decisions.

Objective

- To design a Java-based decision support system for irrigation management.
- To simulate soil moisture, temperature, humidity, and rainfall data for demonstration purposes.
- To implement rule-based logic that determines irrigation requirement and water quantity.
- To generate real-time recommendations and alerts based on simulated conditions.
- To maintain irrigation history and generate summary reports for water usage analysis.
- To demonstrate core Java OOP concepts through a practical agricultural use case.

Problem Statement

- Imagine a small farmer named Ramesh, who owns a few acres of farmland.
- Every morning, he walks through his fields, touches the soil with his hands, looks at the sky, and decides whether his crops need water.
- Some days he irrigates more than necessary, wasting valuable water.
- On other days, he delays irrigation, causing the crops to experience water stress and reducing their yield.
- Ramesh hears about modern IoT-based smart irrigation systems, but the cost of sensors, controllers, and maintenance is far beyond what he can afford.
- Like many small farmers, he continues to rely on experience and guesswork instead of data-driven decisions.

Existing System AND Proposed System

Engineer to Excel

Smart Irrigation Decision Support System Using Simulated Soil Moisture and Weather Data

EXISTING SYSTEM

Traditional irrigation systems rely on manual observation, farmer experience, and fixed schedules. Some advanced systems use IoT sensors, but they require costly hardware and maintenance.

LIMITATIONS

- Relies on manual observation and farmer experience.
- Fixed irrigation schedules cause over/under-irrigation.
- Inefficient water utilization and low productivity.
- Commercial IoT systems require costly sensors & hardware.
- No centralized system for irrigation history and water usage.
- High installation and maintenance cost.
- Limited accessibility for small farmers.

PROPOSED SYSTEM

The proposed system is a software-based decision support system that uses simulated soil moisture and weather data to recommend optimal irrigation. It applies ML algorithms to predict irrigation needs and maintains historical records for analysis.

ADVANTAGES

- Uses simulated soil moisture and weather datasets.
- Eliminates dependency on expensive IoT hardware.
- Provides intelligent irrigation recommendations using ML.
- Improves water-use efficiency and crop productivity.
- Maintains centralized irrigation history and water records.
- Cost-effective, scalable, and easy to implement.
- Can be extended to real-time IoT system in the future.

COMPARISON

Parameter	Existing System	Proposed System
Irrigation Decision	Manual / Experience-based	AI / ML-based Intelligent Recommendation
Data Source	Visual Observation / Fixed Schedule	Simulated Soil Moisture + Weather Data
Irrigation Schedule	Fixed Schedule	Dynamic & Adaptive Schedule
Water Utilization	High Wastage	Optimized Water Utilization
Cost	High (IoT Hardware, Installation, Maintenance)	Low (Software-based, No Hardware)
Irrigation Records	Not Available	Centralized History & Water Usage Records
Accuracy	Low Accuracy	High Accuracy with ML Models
Scalability	Limited	Highly Scalable & Upgradeable
Accessibility	Limited for Small Farmers	Accessible, Affordable & User-friendly

Goal: To develop an intelligent, cost-effective, and accessible irrigation decision support system that optimizes water usage, improves crop yield, and supports sustainable agriculture.

System Architecture

1. User & Farm Management
2. Soil Moisture & Weather Simulation
3. Irrigation Decision Engine
4. Recommendation & Alerts
5. Report Generation & History

Each module hands its output to the next, forming a complete simulate-to-report irrigation pipeline.

Module 1: User & Farm Management

Engineer to Excel

Aspect	Details
Requirements	• User Registration • Login Authentication • Farm Information Storage
Task	• Register farmers • Store farm details • Update farm information
Parameters	User ID, Name, Farm Area, Soil Type, Crop Type, Location
Outcome	• User account created • Farm information stored successfully
Techniques	Java OOP, Encapsulation, Classes, Constructors, Collections

Module 2: Soil Moisture & Weather Data Simulation

Aspect	Details
Requirements	• Soil Moisture Simulator • Weather Generator
Task	• Generate simulated sensor values • Simulate weather conditions
Parameters	Soil Moisture, Temperature, Humidity, Rainfall, Weather Condition
Outcome	• Simulated environmental dataset generated
Techniques	Random Class, Arrays, ArrayList, Java Math Library

Module 3: Irrigation Decision Engine

Aspect	Details
Requirements	• Decision Support Logic
Task	• Analyze simulated data • Decide irrigation requirement
Parameters	Moisture Threshold, Rainfall Prediction, Crop Water Requirement
Outcome	• Irrigation ON/OFF • Required water quantity
Techniques	Rule-Based Logic, If-Else, Java Methods

Module 4: Recommendation & Alerts

Aspect	Details
Requirements	• Recommendation System • Alert Generation
Task	• Recommend irrigation schedule • Notify users
Parameters	Water Amount, Irrigation Time, Weather Alert, Soil Status
Outcome	• Recommendation displayed • Alert messages generated
Techniques	String Handling, Exception Handling, Conditional Statements

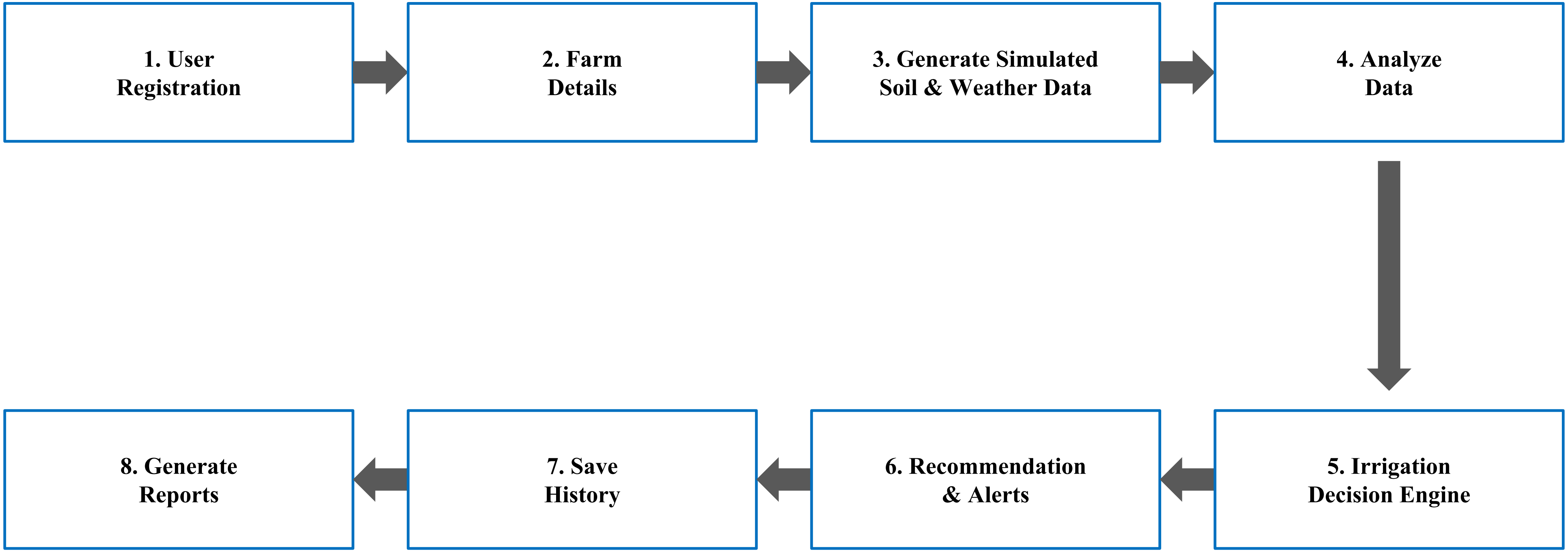
Engineer to Excel

Module 5: Report Generation & History

Aspect	Details
Requirements	• Data Storage • Report Generation
Task	• Store irrigation history • Generate reports
Parameters	Date, Soil Moisture, Weather, Irrigation Decision, Water Usage
Outcome	• Daily history • Summary report • Water usage analysis
Techniques	File Handling, Collections, Searching, Sorting

System Flow

Engineer to Excel



Conclusion

- The Smart Irrigation Decision Support System demonstrates how Java can be used to build a rule-based agricultural decision-making tool.
- Simulated soil moisture and weather data enable a safe, hardware-free demonstration of irrigation logic.
- The system combines data simulation, decision-making, recommendations, and reporting into a single cohesive application.
- It serves as a strong foundation for future integration with real sensors and machine learning models.

Thank You

Engineer to Excel